

Article

Evaluation of The Flexural Properties of Re-entrant Structured Sandwich Beams Fabricated via 3D Printing Using Three-Point Bending Test

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Abstract. A sandwich beam with a re-entrant core has a special negative Poisson's ratio, and this is beneficial for flexural performance (bending stiffness, absorbing energy and so on). However, re-entrant cores have not received much attention despite their unique properties. Comparing this structure to other core designs without prior investigation of its performance in relation to relative density may lead to misleading conclusions about its mechanical behavior. This paper examines the flexural behavior of a 3D-printed re-entrant core sandwich beam with several relative densities in two thermoplastics with different properties: acrylonitrile butadiene styrene (ABS) and polylactic acid (PLA). Three-point bending tests and finite element analysis (FEA) were conducted in this study. Preliminary CO₂ emissions and their prediction equations for ABS and PLA were estimated based on the energy consumed during printing process. The experimental results show that both materials exhibit similar bending characteristics, where bending stiffness increases with relative density, while energy absorption and specific energy absorption (SEA) decrease. A comparison with FEA predictions of bending stiffness and maximum force reveals good agreement, with only a 5–6% difference. Further comparison of materials proves that PLA outperforms ABS in all of flexural properties and they produce less CO₂ emissions. These findings contribute a paradigm for structural optimization and sustainable material selection for next-generation lightweight design.

Keywords: Bending stiffness, CO₂ emissions, Energy consumption, Re-entrant, Sandwich beam

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1. Introduction

Sandwich beam is commonly used in applications, such as automotive, aerospace, and sporting equipment, satellite applications, due to its high stiffness-to-weight ratio and energy absorption capacity [1]. Typical sandwich structures consist of two thin face sheets separated by a lightweight core, which is usually made of polymeric foam as shown in Fig. 1. Currently, sandwich composite structures beam has been continually improved in terms of weight reduction by replacing solid cores with cellular cores such as honeycomb, truss, and circular configurations [2].

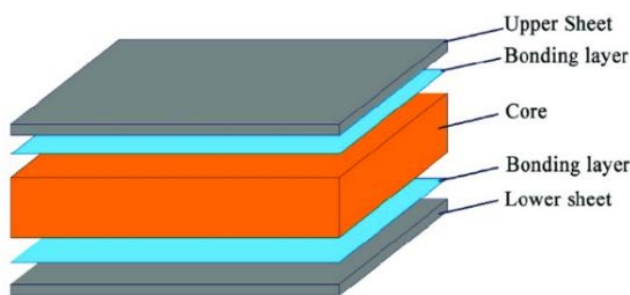


Fig. 1. Components of a sandwich composite structure.

Among the various core architectures, the re-entrant core is a material with distinctive mechanical characteristics and energy absorption capability, characterized by a negative Poisson's ratio [3]. This type of structure, also known as an auxetic material, belongs to a class of mechanical metamaterials that exhibit the unusual behavior of becoming thicker when stretched. The advantages of this architecture include higher energy absorption, indentation resistance, fracture toughness, and improved bending strength [4].

Due to the complexity of the core design, additive manufacturing (AM) has emerged as a suitable fabrication approach for modern production. Known as 3D printing, AM has become popular over the last few decades with applications in construction, packaging, electronics, apparel, as well as healthcare [5]. Many industries have adopted AM as an alternative or complementary manufacturing approach to conventional tooling because of its low cost, flexibility, user-friendliness, and ability to produce intricate components with minimal material waste by joining materials layer upon layer to build up objects when compared with conventional subtractive manufacturing technology [6]. Currently, the AM encompasses various 3D printing technologies to create components, such as Fused Deposition Modelling (FDM), Direct Energy Deposition (DED), and Selective Laser Sintering (SLS), and so on [7] [8]. The FDM technology also known as fused filament fabrication or free-form filament fabrication (FFF), is a commonly utilized approach in 3D printing technologies to create robust, durable, and dimensionally stable parts with accuracy and repeatability, additionally, its hardware and material costs are relatively low. Many applications of FDM have been

reported in the automotive, medical, architectural, and other sectors. This technology is based on material extrusion, creating the desired model by printing layer-by-layer which involves extruding molten thermoplastic or thermoplastic-based filament from an extruder nozzle onto a printing bed [9] [10]. Acrylonitrile Butadiene Styrene (ABS) and Polylactic Acid (PLA) are among the most commonly used thermoplastic filaments in 3D printing applications [11]. ABS is well known for its high toughness, whereas PLA is preferred for its flexibility and ease of processing. In recent years, fiber-reinforced polymers have been widely utilized in the repair, rehabilitation, and strengthening of masonry structures primarily due to their excellent strength-to-weight and stiffness-to-weight ratios, along with their resistance to corrosion [12]. Consequently, their mechanical prowess has received considerable attention from academia and industry alike.

Extensive research has been conducted to better understand the mechanical properties of sandwich beams structures, particularly focusing on the complexity of core designs and topologies. With increasingly demanding service conditions, these structures face a wider range of potential damage mechanisms under various loading scenarios, including axial compression, localized impacts, and bending. Therefore, evaluating the strength and stiffness of sandwich beams under bending loads is essential. In the current research, most studies investigate the mechanical performance of non-auxetic cores compared to auxetic cores type, or focus on modifications of conventional re-entrant cores by controlling specimen dimensions and using three-point bending tests. N.K. Choudhry et al. [13] studied both conventional and modified re-entrant cores to enhance mechanical properties. The results show that the modified core exhibits higher bending stiffness and energy absorption than the conventional design. M. Najafi et al. [14] investigated auxetic and non-auxetic cores to compare their mechanical performance. Their results indicate that the arrowhead core (auxetic) has the highest bending stiffness, while the honeycomb core exhibits the highest energy absorption. However, studies specifically focusing on re-entrant cores remain limited. Comparing a re-entrant core without prior investigation into the effects of relative density, which directly influences bending stiffness and energy absorption with other core structures may result in misinterpretation of its mechanical behavior. When such comparisons are made alongside results from other core architectures, it becomes difficult to accurately identify which design provides superior mechanical performance.

On the other hand, the study of energy consumption and CO₂ emissions in AM has gained increasing attention since the industry sector is a significant contributor to global CO₂ emissions [15]. To minimize the effects of climate change, many researchers have begun studying 3D printing, including its material usage, from an environmental impact perspective [16]. With sustainability becoming a priority in manufacturing, numerous researchers have examined factors in AM that contribute

to greenhouse gas (GHG) emissions. Among these, energy consumption stands out as a major contributor, representing a significant portion of the industry's overall energy use. Osman Ulkir. [17] studied energy consumption based LCA using the STL process on PLA and ABS materials. The result show that PLA and ABS CO₂ emission from manufacturing phase are produced around 15% to 20% of the entire life cycle. Guanyi Chen et al. [18] studied the impact of CO₂ emissions from replacing traditional plastics with biodegradable plastics and showed that product manufacturing phase contribute CO₂ emission of around 18-30% of the entire cycle depending on end-of-life options. Although its impact appears minor when considering the entire cycle, it accounts for approximately one-third of CO₂ emissions during the manufacturing phase. Accordingly, many researchers have investigated AM factors which contribute to increased energy consumption whether varying material, AM setting (Nozzle temperature, speed, layer height, and others) or 3D printing method. Shubhada Wark and V.S. Puranik [19] studied on energy consumption of ABS and PLA with FDM process. The printed shape, printing speed and infill were controlled. The result show that ABS printing energy consumption for all experiments is almost 1.5 to 2 times higher than that of printing with PLA.

Although several studies have been reported, gaps remain regarding the nonlinear behavior and failure mechanisms being examined. Also, it is uncertain if the behavior of sandwich beams is dominated by the properties of the material or changed depending on the relative density. As a result, these aspects warrant further investigation. Therefore, to provide a more comprehensive understanding, this study analyzes the flexural properties of re-entrant structures fabricated from ABS and PLA materials with relative densities of 0.3, 0.4, and 0.5. Both experimental testing and FEA were employed to examine the mechanical behavior of sandwich beam structures up to the nonlinear region, including the investigation of failure mechanisms. The study began with the characterization of the mechanical properties of ABS and PLA materials, which were tested according to ASTM D638 standards. The sandwich beam specimens were manufactured following ASTM C393 standards. In addition, a preliminary estimation of CO₂ emissions from the manufacturing process was conducted based on electrical energy consumption during the AM process. Production time and energy usage were measured using a stopwatch and an electrical power meter.

This study presents a further investigation into the nonlinear behavior, damage mechanisms, and dominant deformation characteristics of re-entrant core sandwich structures under three-point bending. This approach enhances the understanding of structural responses and deformation mechanisms, leading to improved structural design and more reliable failure prediction. In addition to mechanical performance, a preliminary environmental assessment was conducted by estimating CO₂ emissions from electricity consumption during the additive

manufacturing (AM) process. The main contribution of this work is the development of an integrated framework linking mechanical performance, damage mechanisms, relative density, and environmental impact to support optimized sandwich structure design in terms of mechanical efficiency and sustainability.

2. Materials and Method

This section presents the methodology used to investigate the flexural behavior of 3D-printed sandwich beams with re-entrant cores. The study involves designing core geometries at various relative densities, fabricating specimens using ABS and PLA, performing tensile and three-point bending tests, and validating results through FEA.

2.1. Re-entrant core design

The geometric features of the proposed sandwich core design based on a re-entrant configuration are presented in this section. Fig. 2. illustrates the schematic representation of the re-entrant microstructure, detailing the characteristic geometry such as cell angle, wall thickness, strut length, and unit cell height. The core design follows initial relative density 0.3 and varying to 0.4 and 0.5 under different configurations calculated based on Eq. (1) and the in-plane Poisson's ratio can be evaluated Eq. (2).

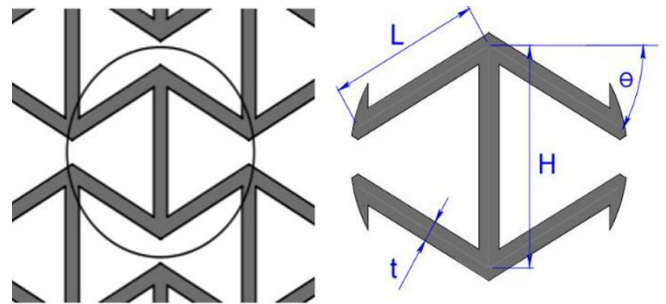


Fig. 2. Schematic of re-entrant structures.

$$\frac{\rho^*}{\rho_s} = \frac{(t/L)[(H/L)+2]}{2\cos\theta[(H/L)+\sin\theta]} \quad (1)$$

$$\nu_{12} = \frac{\cos^2\theta}{[(H/L)+\sin\theta](\sin\theta)} \quad (2)$$

For re-entrant, θ is negative, and in 2D isotropic materials exhibit a broader allowable range for Poisson's ratio, extending from -1 to 1 [20]. In the case of engineered structures like re-entrant core beams, which leverage geometric design rather than intrinsic material properties, the Poisson's ratio can reach values as low as -0.7 to -0.8 [21].

In this study, all specimens were designed with dimensions of 150 mm × 40 mm × 30 mm using an NPR -0.75 and consist of 14 × 2 unit cells. The relative densities

were varied at 0.3, 0.4, and 0.5. Based on Eq. (1) and (2), the corresponding cell dimensions are presented in Table 1 and Table 2. To ensure uniform load distribution, two thin face sheets with a thickness of 1 mm are added to the top and bottom of the core structure. Fig. 3 illustrates the sandwich beam used for the bending test.

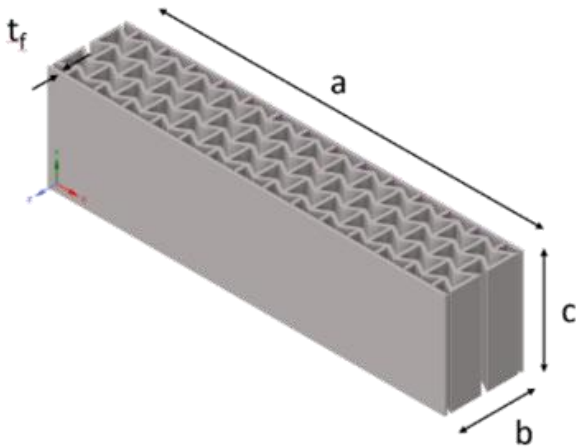


Fig. 3. Sandwich beam specimen. Here, a is the beam length; b is the core thickness; c is the beam width; t_f is the face sheet thickness.

Table 1. Core design parameters of re-entrant structure

Relative density	0.3	0.4	0.5
a (mm)	150.53	150.80	151.03
b (mm)	39.56	39.65	39.67
c (mm)	39.63	29.60	29.40
t_f (mm)	← 1 →		

Table 2. Specimen design parameters

$\frac{\rho^*}{\rho_s}$	L (mm)	H (mm)	t (mm)	θ	Number of cells
0.3	6.20	10.11	1.00	30°	14 (2layers)
0.4			1.34		
0.5			1.67		

The re-entrant core structures were fabricated FDM 3D printer (Bambu Lab P1S) with ABS and PLA materials. The layer-by-layer deposition process inherently introduces anisotropic behavior, as interlayer bonding strength is generally lower than intralayer strength. Consequently, printing orientation significantly influences mechanical properties. However, constructing geometry as a layer-by-layer in model for fully replicating the actual printed remains a limitation of the present modelling framework. This restriction makes the direct implementation of anisotropic material behavior difficult within the current model formulation. Therefore, the material was assumed to exhibit isotropic behavior, which

is commonly adopted to reduce model complexity while providing reasonable agreement with experimental results. To justify the isotropic assumption, the specimens were designed to exhibit near-isotropic responses as much as possible. All samples were fabricated with solid structures (100% infill) under identical processing conditions and the same build orientation. The face sheets were aligned unidirectionally along the 0° direction, while the core structures followed their geometric orientation. These conditions provided sufficiently uniform mechanical responses, allowing isotropic material behavior to be assumed in subsequent analyses, as illustrated in Fig. 4.

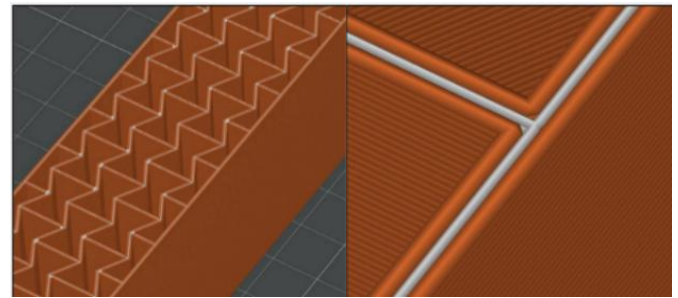


Fig. 4. Printing orientation of the specimens.

Moreover, energy consumption and CO₂ emission were measured and calculated in Eq. (3) and Eq. (4).

$$E = \frac{P \times \text{time}}{1000} \quad (3)$$

$$C_e = E \times E_f \quad (4)$$

When E : energy consumption (Kwh), P : power (Kw), C_e : carbon emission (kg CO₂e) and E_f : emission factor (kg CO₂e / Kwh) which is 0.4401 kg CO₂e / Kwh based on Thailand Greenhouse Gas Management Organization [22].

2.2. Materials properties

To establish baseline material properties for ABS and PLA, standard dog-bone specimens were printed using identical parameters, with three samples for each material, and tested in uniaxial tension according to ASTM D638, as shown in Fig. 5(a). This enabled reliable measurement of tensile strength, Young's modulus, and strain at break, which are essential for validating both experiments and FEA. The tensile test setup is shown in Fig. 5(b), and the resulting stress-strain curves are presented in Fig. 6. The extracted properties are summarized in Table 3.

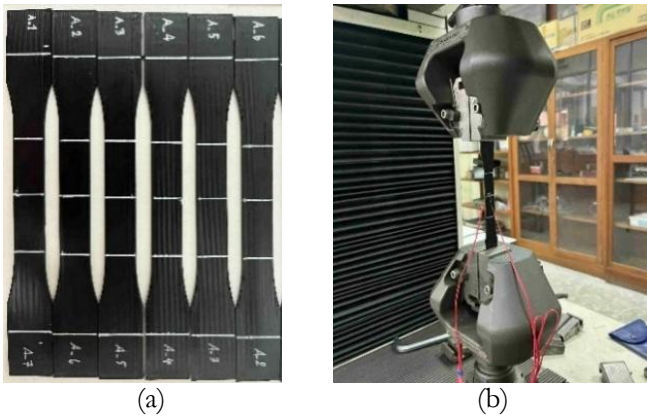


Fig. 5. (a) Example for printed dog bone and (b) uniaxial tensile test setup and testing.

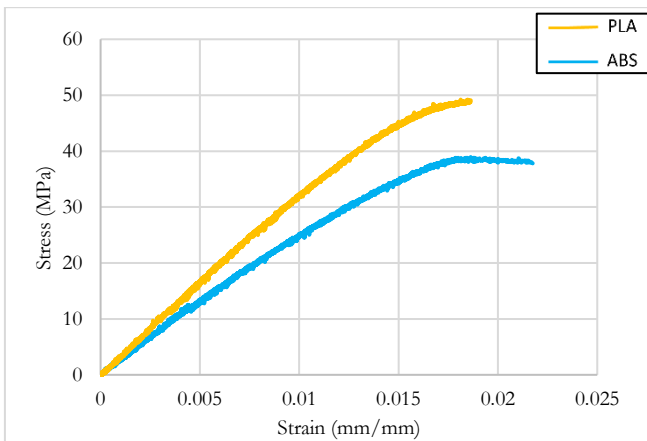


Fig. 6. Comparison of engineering stress–strain response of ABS and PLA at 0°.

Table 3. Mechanical properties of ABS and PLA at 0°

Item/Material	ABS	PLA
Density (kg/m ³)	1050	1250
Young's Modulus (MPa)	2471	3339
Poisson's Ratio	0.35	0.3
Tensile Yield Strength (MPa)	38.14	44.22

2.3. Three point bending test setup

The three-point bending test was performed using an Autograph AG-IS 100KN electronic universal testing machine. The program setting follows the ASTM C393 standard with quasi-static loading rate 6 mm/min and the span length was 100 mm as shown in Fig. 7.

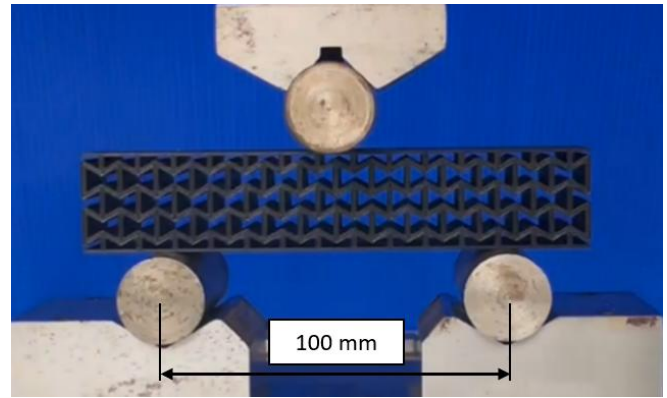


Fig. 7. Three-points bending test and specimen setup.

The outcomes of the bending tests are presented as force–deflection curves. The slope of these curves represents the bending stiffness, as defined in Eq. (5), where F denotes the applied force (N), δ represents the deformation in the direction of the applied force (mm), and K_b is the stiffness (N/mm). Furthermore, the energy absorption (EA, kJ) and the specific energy absorption (SEA, kJ/kg) can be derived from these curves according to Eq. (6) and Eq. (7) [23].

$$K_b = \frac{F}{\delta} \quad (5)$$

$$EA(s) = \int_0^{\delta} F(\delta) d\delta \quad (6)$$

$$SEA = \frac{EA}{m} = \frac{\int_0^{\delta} F(\delta) d\delta}{m} \quad (7)$$

2.4. Numerical modelling

FEA was employed to investigate the flexural behavior of the re-entrant structure using material properties of ABS and PLA obtained from experimental testing and implemented through engineering data. The numerical simulation of the three-point bending test was conducted in ANSYS using the Static Structural module, considering nonlinear effects arising from large deformation and frictional contact. The material behavior was assumed to be perfectly elastic–plastic. Fig. 8 illustrates the mesh configuration (0.5 mm) and boundary conditions applied to the re-entrant core structure. The model used a 0.5 mm mesh with linear multizone hexahedral 8-node brick elements (SOLID185). A friction coefficient of 0.25 was defined between the rollers and face sheets. The load was applied vertically at the indenter, while fixed supports were assigned at both support locations. The deflection of the sandwich beam and corresponding von Mises stress distribution were obtained under the prescribed conditions. Furthermore, the Large Deformation option in the analysis settings was selected to consider geometric nonlinearity of large structural deformations. However, the numerical model implemented in this work does have its limitations

regarding the simulation of structural buckling behaviour. Although solid elements can capture stress gradients through the thickness, in bending-dominated conditions the formulation may potentially be affected by shear locking, which can introduce additional numerical stiffness under certain bending-dominated conditions. Moreover, geometric imperfections were not incorporated, as the model assumes an idealized geometry without initial defects and eigenvalue buckling analysis was not included in the present, and further clarification is provided in section 3.1.

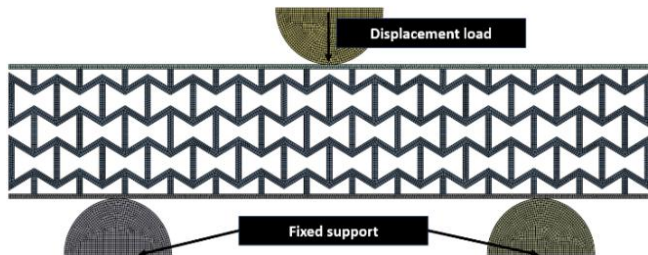


Fig. 8. Mesh configuration (0.5 mm element size) and boundary conditions.

3. Results and discussion

This section presents experimental and numerical results of sandwich beams with re-entrant cores, focusing on the effects of relative density and material type on bending performance, including stiffness and SEA. FEA results are compared with experiments to validate the model. Failure mechanisms during three-point bending are analysed, and the energy consumption and associated CO₂ emissions from the 3D printing process are estimated to assess environmental impact.

3.1. Bending performance of re-entrant core

Figures 9 and 10 present the force–deflection responses of ABS and PLA specimens at different relative densities. In the linear elastic stage, the highest bending stiffness is observed at a relative density of 0.5, followed by 0.4 and 0.3. During the yielding stage, specimens with relative densities of 0.4 and 0.5 reach the maximum force prior to failure, whereas the 0.3 specimen attains its peak force after yielding and continues deforming up to a deflection of 30 mm due to testing limitations. The highest EA and SEA occur at a relative density of 0.3, as low-density structures absorb energy mainly through cell wall bending, resulting in greater ductility. In contrast, higher-density structures exhibit greater stiffness and strength but fail in a more brittle manner at stress concentration regions.

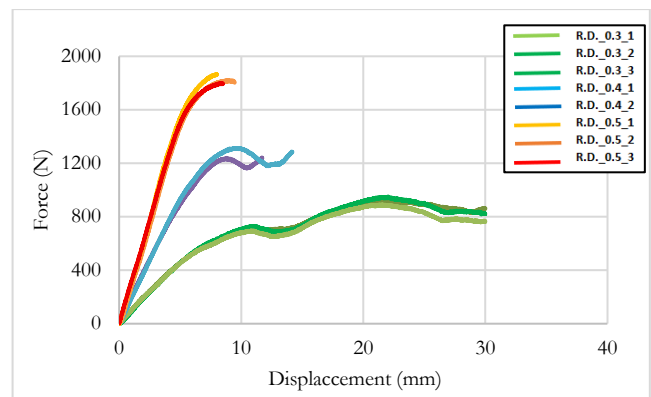


Fig. 9. Force–deflection plot obtained from experiments on ABS specimens at relative densities of 0.3, 0.4, and 0.5.

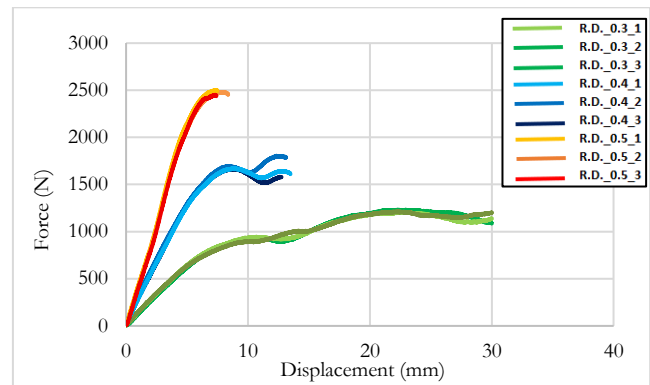


Fig. 10. Force–deflection plot obtained from experiments on PLA specimens at relative densities of 0.3, 0.4, and 0.5.

When comparing the mechanical performance of ABS and PLA, both materials exhibit similar response trends across all relative densities, indicating that structural behavior is primarily governed by the geometry of the re-entrant core. PLA demonstrates approximately 40% higher bending stiffness and about 33% higher maximum load than ABS at all relative density levels, attributed to its higher Young's modulus,

At low relative densities, energy absorption is primarily governed by cell wall flexibility. The observed deformation is dominated by cell wall bending and plastic hinge formation rather than buckling, particularly at a relative density of 0.3. The load–deflection curve at this density does not show the sudden drop of stiffness typically associated with buckling, indicating progressive plastic deformation prior to failure. Although the deformation modes are similar for both materials, PLA demonstrates greater energy absorption owing to its higher modulus. The difference in energy absorption between PLA and ABS decreases from 74% at low relative density to 38% and 20% at medium and high densities, respectively. Likewise, SEA differences reduce from 42% to 13% and eventually converge to nearly 0%. These results suggest that material effects dominate at low relative densities, whereas structural geometry becomes more influential as relative density increases. Comparative results are presented in Fig. 11(a)–(d).

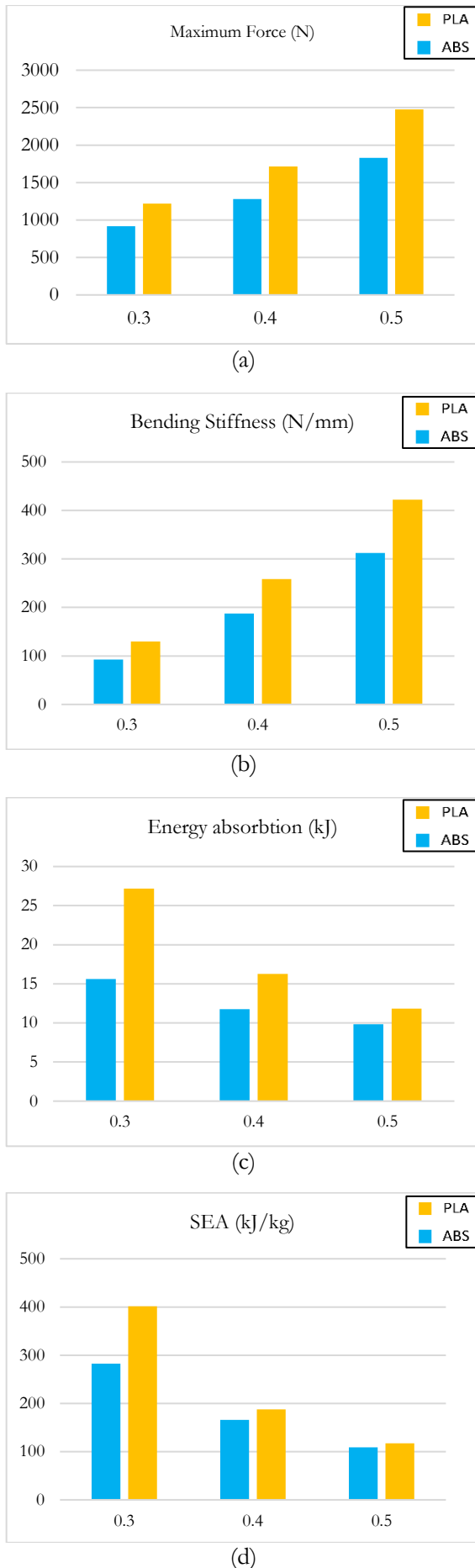
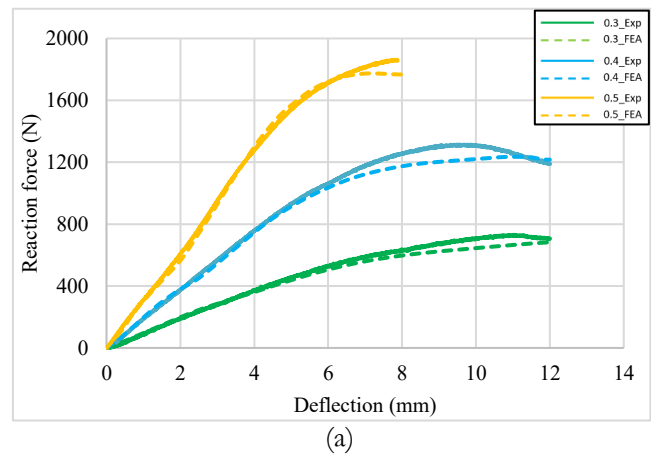


Fig. 11. Comparison of flexural properties of ABS and PLA: (a) maximum force, (b) bending stiffness, (c) energy absorption (EA), and (d) specific energy absorption (SEA).

3.2. Finite element result

FEA was employed to predict bending stiffness and maximum load based on the force–deflection curve. The displacement range was limited to 0–12 mm, corresponding to the experimentally observed failure for relative densities of 0.4 and 0.5. For the relative density of 0.3, the model deflection could not reach 30 mm due to localized structural instability within the core, causing deformation incompatibility. Figures 12(a) and 12(b) compare force–deflection curves between experiments and FEA predictions for ABS and PLA. The maximum load and bending stiffness for relative densities of 0.3 (at yield), 0.4, and 0.5 are summarized in Tables 5 and 6. The simulations show good agreement with the experiments, with differences of 1–5% for ABS and 2–10% for PLA. Discrepancies may be attributed to specimen imperfections such as voids, which reduce the effective load-carrying area and increasing stress levels in the remaining material. Under bending-dominated behavior, voids may slightly reduce the effective second moment of area, resulting in greater deformation. Furthermore, void boundaries act as geometric discontinuities that induce stress concentration and facilitate damage initiation. Voids also disrupt load transfer paths, causing stress redistribution and reducing global stiffness and load carrying capacity. Fig. 13 illustrates that void formation may result from extrusion inconsistencies, filament quality variations, or nozzle pathing during printing, creating small gaps and reducing specimen weight.



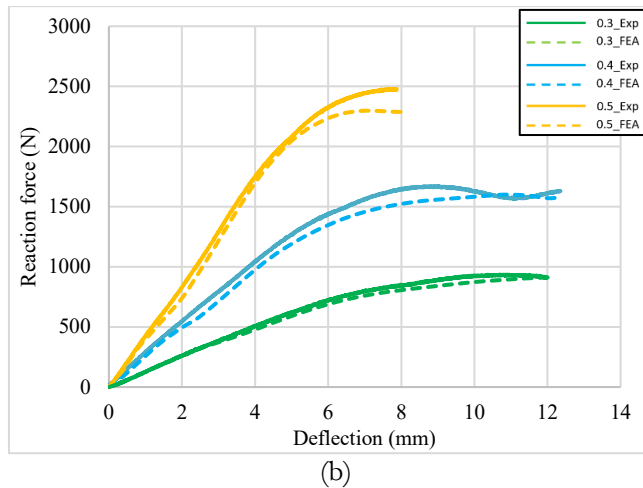


Fig. 12. Comparison of bending behavior between experimental results and FEA predictions for both materials: (a) ABS and (b) PLA.

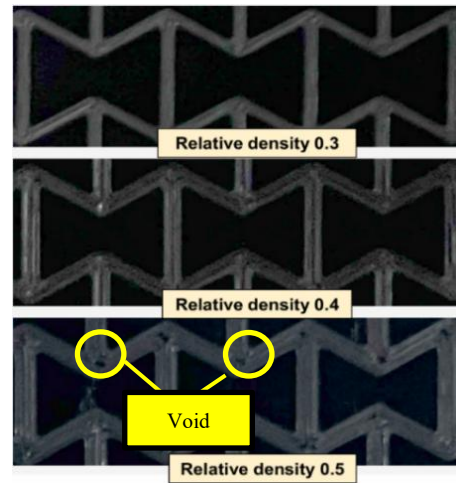


Fig. 13. Imperfections of printed specimens at relative densities of 0.3, 0.4, and 0.5.

Table 4. ABS and PLA maximum force, bending stiffness, energy absorption and SEA under bending test with relative density 0.3, 0.4 and 0.5

Materials	Relative density	Maximum Force (N)	Bending stiffness (N/mm)	Energy absorption (kJ)	SEA (kJ/kg)
ABS	0.3	918.75	92.61	15.60	282.41
	0.4	1280.47	187.21	11.77	166.00
	0.5	1828.65	312.12	9.83	108.90
PLA	0.3	1220.31	129.92	27.19	401.25
	0.4	1713.54	258.32	16.26	187.78
	0.5	2476.56	422.14	11.83	117.04

Table 5. Comparison maximum load and bending stiffness between experiment and FEA of ABS

Relative density	Exp		FEA		Differences (percentage)	
	Max Force (N)	Bending stiffness (N/mm)	Max Force (N)	Bending stiffness (N/mm)	Max Force	Bending stiffness
0.3	731.25	92.61	752.23	96.33	2.87	4.03
0.4	1314.06	187.21	1366.20	189.68	3.97	1.32
0.5	1867.20	312.12	1955.40	310.28	4.72	0.59

Table 6. Comparison maximum load and bending stiffness between experiment and FEA of PLA

Relative density	Exp		FEA		Differences (percentage)	
	Max Force (N)	Bending stiffness (N/mm)	Max Force (N)	Bending stiffness (N/mm)	Max Force	Bending stiffness
0.3	1220.31	129.92	911.54	123.64	5.29	4.83
0.4	1713.54	258.32	1815.00	242.02	8.66	6.31
0.5	2476.56	422.14	2297.90	397.60	7.33	5.81

3.3. Failure mechanism analysis

From experimental observations, cracking initiates at the bottom surface of the midspan face sheet and propagates into the core. Von Mises stress analysis confirms that the maximum stress concentration occurs at the bottom face sheet, consistent with the observed crack initiation. Figures 14(a)-14(c) compare the numerical and experimental failure modes for relative densities of 0.3, 0.4, and 0.5, showing good agreement. For relative densities of

0.4 and 0.5, failure is governed by face sheet tension at the midspan, triggered by core yielding and culminating in face sheet fracture, which defines the ultimate bending capacity of the sandwich beam. In contrast, the 0.3 specimen remains unbroken even at large deflections (up to 30 mm). The lower-density core provides greater structural compliance, promoting progressive plastic collapse rather than brittle fracture, allowing continued deformation and energy absorption. Both PLA and ABS exhibit similar failure sequences across all relative densities, indicating

that the response is primarily controlled by the re-entrant cellular geometry. Although material stiffness affects stress magnitude, stress concentration at re-entrant corners and face sheet–core interaction governs the failure progression. Agreement between experimental and FEA results at the final deflection (for relative densities of 0.4 and 0.5) further validates the model. However, reaching the yield stress does not necessarily indicate final fracture, as demonstrated by the 0.3 specimens, which continue deforming beyond 12 mm due to progressive core collapse. The same trend is observed for PLA.

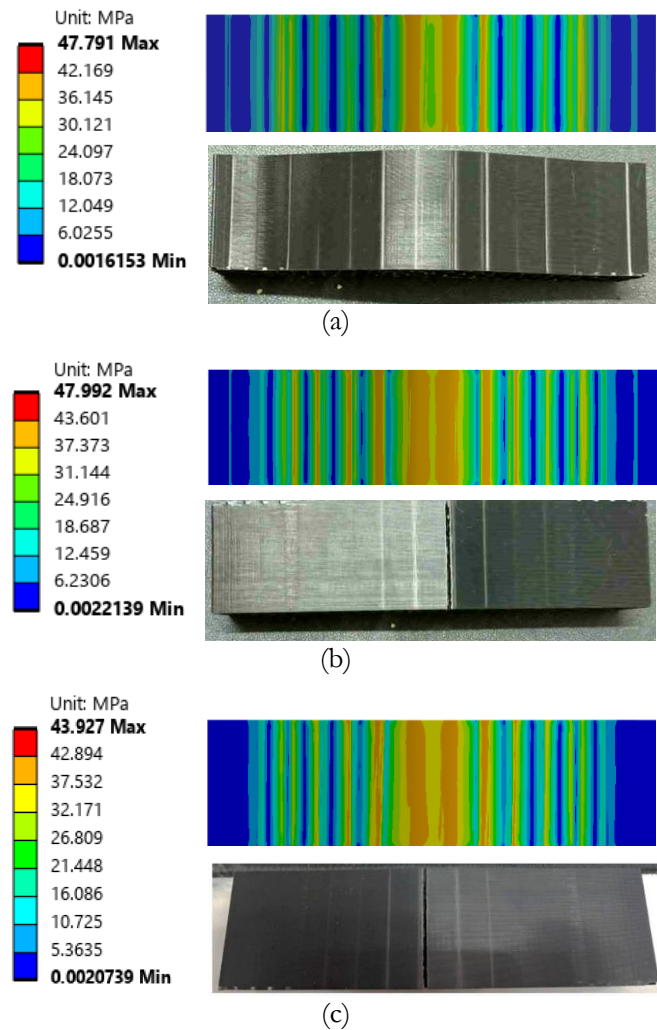


Fig. 14. Comparison of experimental results and FEA predictions for failure mechanisms of ABS at relative densities of (a) 0.3, (b) 0.4, and (c) 0.5, showing the equivalent stress distribution in the bottom face sheet.

Table 7. Comparison of deflection and maximum stress in the bottom face sheet of ABS sandwich structures between experimental and FEA results at different relative densities

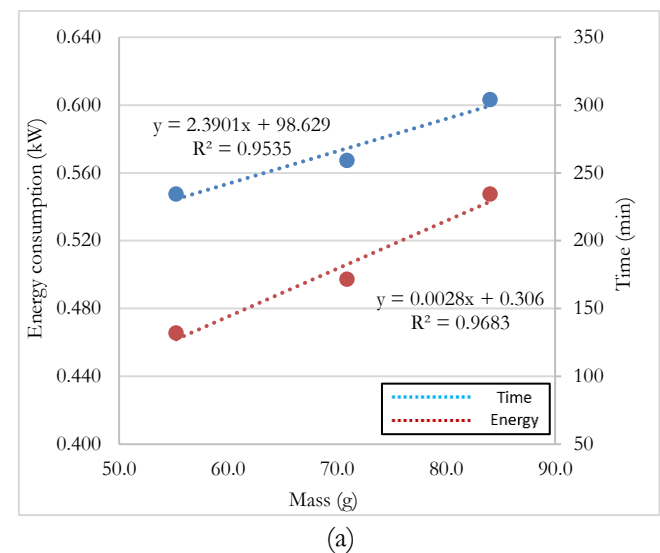
Relative density	Exp		FEA	
	Deflection (mm)	Stress (MPa)	Deflection (mm)	Stress (MPa)
0.3	12	38	12	37-40
0.4	12	38	12	37-38
0.5	8	38	8	37-38

Table 8. Comparison of deflection and maximum stress in the bottom face sheet of PLA sandwich structures between experimental and FEA results at different relative densities

Relative density	Exp		FEA	
	Deflection (mm)	Stress (MPa)	Deflection (mm)	Stress (MPa)
0.3	12	44	12	42-46
0.4	12	44	12	42-44
0.5	8	44	8	42-44

3.4. Energy consumption and CO₂ emissions analysis

The analysis of energy consumption and CO₂ emissions corresponding to relative densities of 0.3, 0.4, and 0.5 for both ABS and PLA during the manufacturing process is presented in this study. The relationships among printed mass, energy consumption, and printing time are illustrated in Fig. 15(a) and Fig. 15(b). Energy consumption increases consistently with printed mass, showing increments of approximately 6–10% for ABS and 9–17% for PLA. A linear correlation between printed mass and energy consumption was observed. Using an emission factor of 0.4401 kg CO₂e/kWh from the Thailand Greenhouse Gas Management Organization, CO₂ emissions were calculated and expressed in Eqs. (7) and (8), where m_{ABS} and m_{PLA} represent mass (g) and Y_{CO_2} denotes emissions (kg CO₂e).



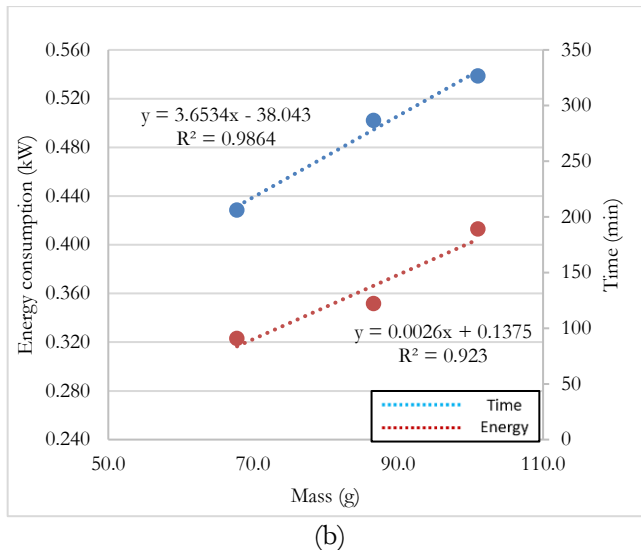


Fig. 15. Mass vs energy consumption and time for both materials: (a) ABS and (b) PLA

$$Y_{CO_2} = (1.23 \times 10^{-3}) m_{ABS} + 0.135 \quad (7)$$

$$Y_{CO_2} = 1.14 \times 10^{-3} m_{PLA} + 0.060 \quad (8)$$

Figure 16 presents the corresponding emissions across the three relative densities. PLA exhibits significantly lower CO₂ emissions than ABS due to its lower extrusion temperature (~190–220 °C) compared with ABS (~220–260 °C). Additionally, PLA is a bio-based and biodegradable polymer derived from renewable resources such as corn starch or sugarcane. However, it is relatively brittle and less suitable for high-temperature or long-term applications. In contrast, ABS is petroleum-based, non-biodegradable, and requires higher processing temperatures, resulting in greater energy demand and emissions per unit mass. Nevertheless, ABS provides superior toughness, impact resistance, and durability, making it more suitable for structural applications requiring ductility. Overall, this trade-off highlights the importance of selecting materials in AM according to application requirements, balancing sustainability and structural performance. Environmental impact is also influenced by processing parameters, including build orientation, infill density, and support structures. Process optimization strategies—such as minimizing supports, optimizing infill, and applying adaptive layer heights—can significantly reduce energy consumption and emissions. Therefore, achieving sustainability in additive manufacturing requires both appropriate material selection and optimized design and processing strategies.

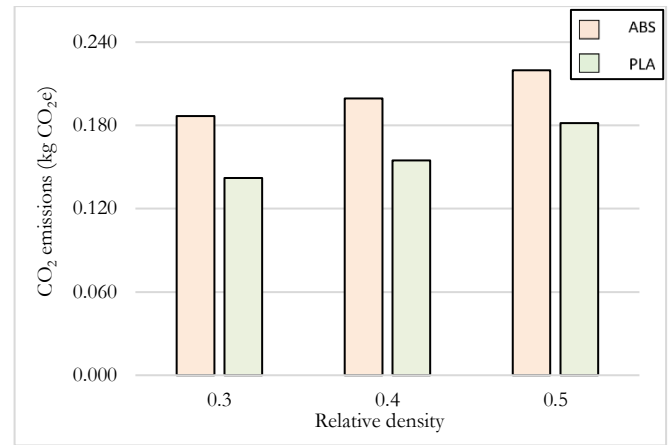


Fig. 16. Comparison of CO₂ emissions between ABS and PLA.

4. Conclusions

In this study, re-entrant cores were investigated to evaluate their mechanical performance using two different materials (ABS and PLA) through both experimental testing and FEA. The strong agreement between experimental and numerical results up to the yield point supports using the calibrated models to predict bending stiffness and maximum load across the studied relative densities. As expected, higher relative densities provided greater bending stiffness but at the expense of reduced energy absorption. A direct material comparison revealed that PLA outperformed ABS in terms of flexural properties while also consuming less energy. Furthermore, preliminary equations were developed to predict CO₂ emissions for both materials. Importantly, the predictive framework for stiffness, maximum load, and emissions is transferable to other printable materials by substituting appropriate constitutive properties and process parameters (e.g., PETG, nylon, or fiber-reinforced polymers). These results indicate that higher relative densities are suitable for stiffness-dominated applications, whereas lower relative densities favor energy absorption. Integrating mechanical performance with environmental indicators further demonstrates the potential for sustainable lightweight structural design. However, this study still has several limitations that may provide directions for future research. First, the geometry used in the numerical model assumes isotropic material behavior due to the limitations associated with explicitly modeling the layer-by-layer deposition process. Consequently, the model may not fully capture the anisotropic characteristics introduced by additive manufacturing. Second, internal voids inherent to the printing process were not incorporated into the numerical model. Owing to these assumptions, the predicted stiffness and strength may slightly deviate from the actual material behavior, as discussed in Sections 2.1 and 3.3.

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